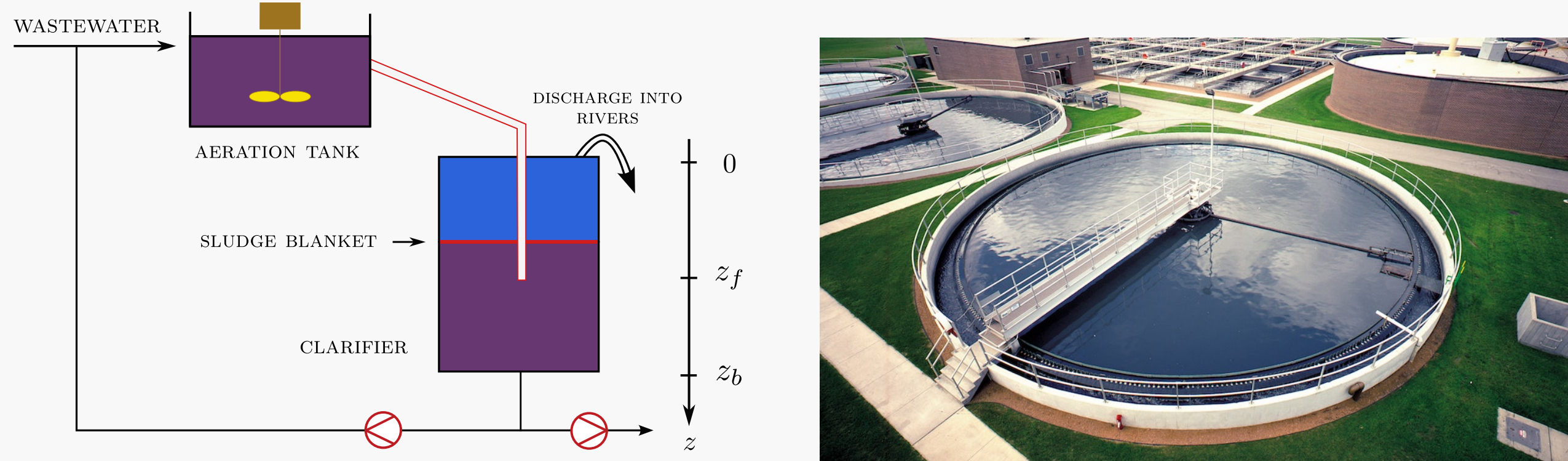


INTRODUCTION AND ENGINEERING CONTEXT



1D-DYNAMIC MODEL OF THE CLARIFIER (TWO-PHASE FLOW)

Let $(t, z) \in \mathbb{R}^+ \times [0, z_b]$.

Consider *conservative variables*: ε_s (Particles density), $f_s := \varepsilon_s v_s$ (Momentum) with v_s the Particles velocity

$$\underbrace{\frac{\partial}{\partial t} \begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \underbrace{\frac{\partial}{\partial z} \begin{pmatrix} f_s \\ \varepsilon_s^2 + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \end{pmatrix}}_{\mathbf{F}(\mathbf{Y})} = \underbrace{\begin{pmatrix} \frac{f_{1s}}{\rho_s} \delta_{z_f} \\ \varepsilon_s g \left(1 - \frac{\rho_l}{\rho_s}\right) + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \frac{f_{2s}}{\rho_s} \delta_{z_f} \end{pmatrix}}_{\mathbf{S}(\mathbf{Y})} \quad (1)$$

with a pressure function: $\sigma_e(\varepsilon_s) = \begin{cases} 0, & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \frac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}, & \varepsilon_s > \varepsilon_c \end{cases}$

δ_{z_f} is a Dirac distribution located at $z = z_f$

$r(\varepsilon_s)$ is the liquid/solid drag force

v_m is the average velocity of the liquid/solid mixture

f_{1s}, f_{2s} represents the sludge feed inlet.

ρ_l : water density / g : gravity acceleration constant

$\sigma_0, n_s, n_r, \varepsilon_c, \rho_s$ (set of parameters)

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices (pumps).

$$f_s(0, t) = U_0(t), \quad f_s(z_b, t) = U_1(t)$$

U_0, U_1 are *signal commands*.

GOALS

- Design a robust boundary controller that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth
- Prove the stability of the controlled system

METHODOLOGY AND LINEARIZED MODEL

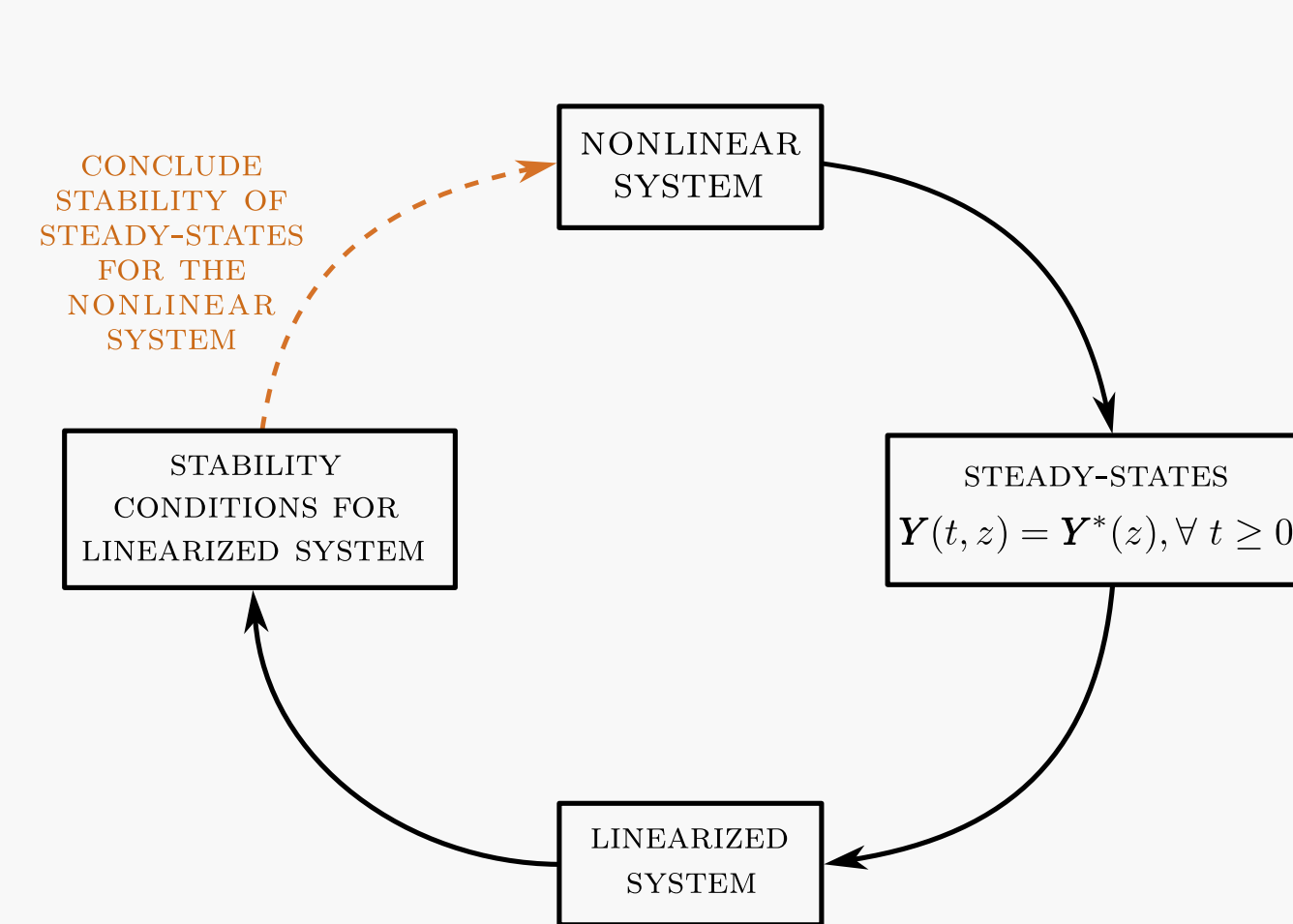


Figure 1. Roadmap on how to stabilize

INFORMATION

In the existing literature the issue of stabilizing *strictly hyperbolic* systems is widely addressed (see Bastin and Coron 2016, Bressan 2000) but only few results and even fewer control designs exists for *weakly hyperbolic* systems, such systems emerge in various physical areas (see Keyfitz and Kranzer 1980, Amirat et al. 2024)

The Linearized System take the following general form in Riemann coordinates (we assume dynamic boundary conditions):

$$\begin{cases} \partial_t \phi - A(x) \partial_x \phi = B(x) \phi, \\ \frac{d}{dt} \phi(t, 1) = D \phi(t, 0) + E \phi(t, 1) + u, \\ \phi(0, x) = \phi_0(x), \quad x \in (0, 1). \end{cases} \quad (2)$$

$\phi : [0, +\infty) \times [0, 1] \rightarrow \mathbb{R}^2$ is the state, $A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}$ where we assume that $c(x) \neq 0, \forall x \in [0, 1]$ and $\lambda_{1,2} > 0$ such that there exists some $x \in [0, 1]$ for which $\lambda_1(x) = \lambda_2(x)$.

$B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}$, u is the control input, $D, E \in \mathcal{M}_{2 \times 2}(\mathbb{R})$.

- The coefficient matrix $-A(x)$ is not diagonalizable throughout the entire spatial domain, classifying system (2) as *weakly hyperbolic*.

CONTROL DESIGN

Select a control law given by

$$u = -(\kappa \mathbb{I}_2 + E) \phi(1), \quad \kappa > 0 \quad (3)$$

STABILITY ANALYSIS OF THE CLOSED-LOOP SYSTEM

Assumption: There exists a function $\mathfrak{b} : [0, 1] \mapsto \mathbb{R}$ such that $B_{21}(x) = \mathfrak{b}(x)(\lambda_2(x) - \lambda_1(x))$.

THEOREM:

Under above Assumption and for sufficiently small $B(x)$ and D , we have that System (2)-(3) is exponentially stable for the $L^2 \times H^1 \times \mathbb{R}^2$ -norm, $\forall \kappa > 0$.

The proof relies on constructing the following *non-quadratic* Lyapunov functional:

$$\mathcal{V}(\phi) = \int_0^1 \frac{1 + \mu x}{\lambda_1(x)} \phi_1^2 dx + \int_0^1 \frac{1 + \mu x}{\lambda_2(x)} \phi_2^2 dx + \alpha \int_0^1 \frac{1 + \mu x}{\lambda_2(x)} (\partial_x \phi_2)^2 dx + 2\alpha \int_0^1 (1 + \mu x) \mathfrak{b}(x) \phi_1 \partial_x \phi_2 dx + \eta w^\top w$$

with $\mu, \alpha, \eta > 0$ and $w = \begin{pmatrix} \phi(t, 0) \\ \phi(t, 1) \end{pmatrix}$.

NUMERICAL RESULTS AND DISCUSSIONS

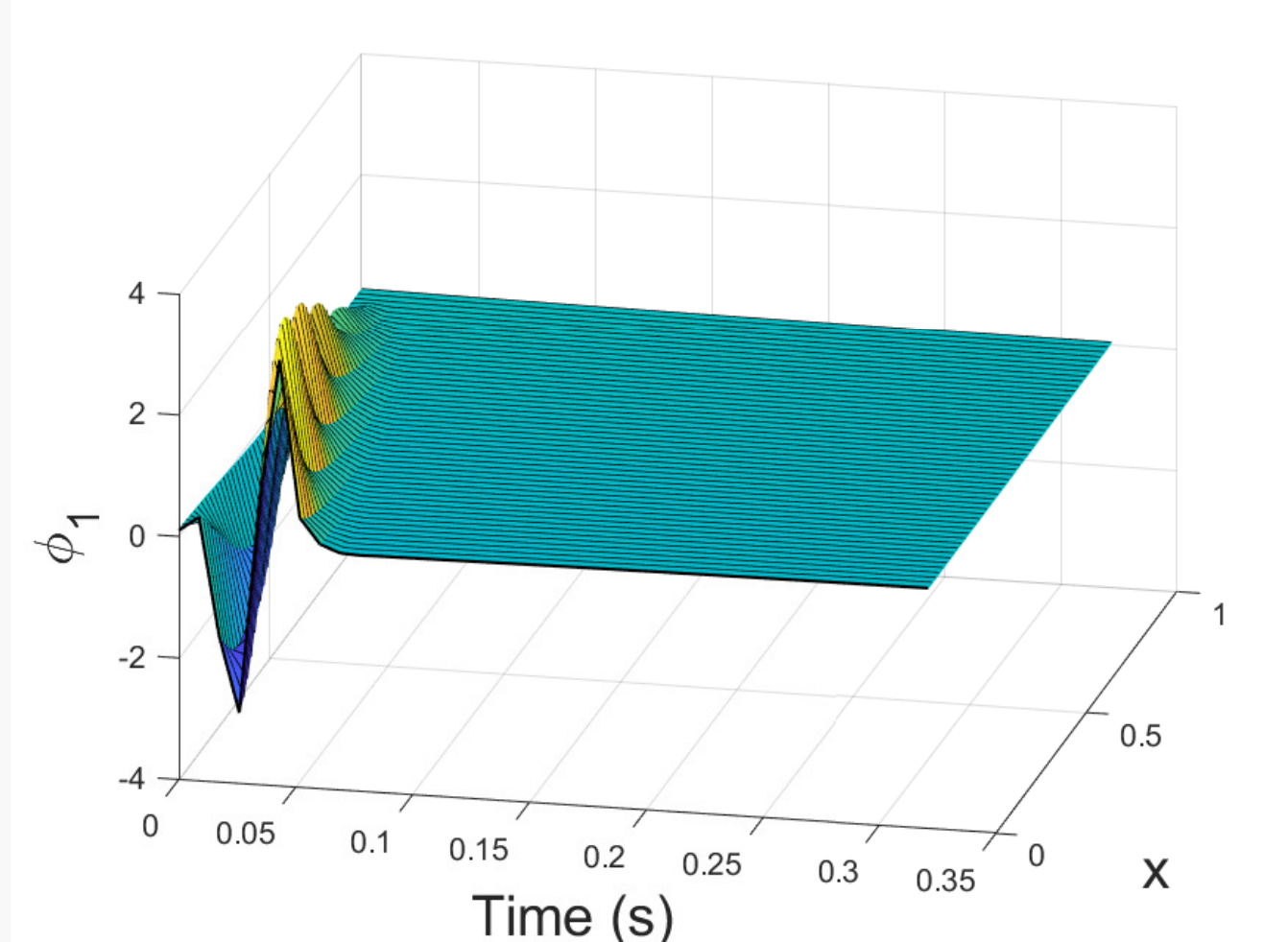


Figure 2. Time-Space evolution of the state-variable ϕ_1

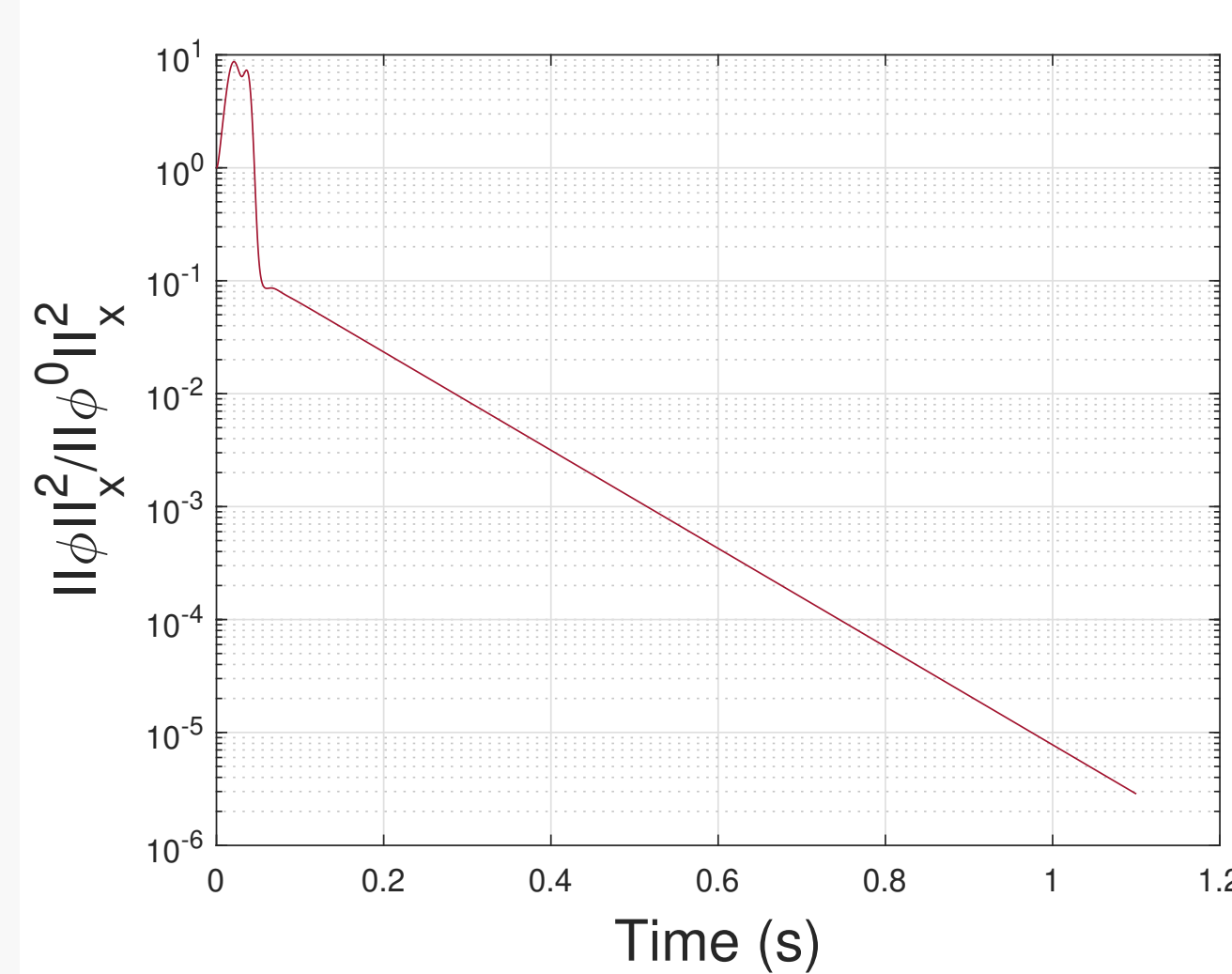


Figure 3. Time evolution of $\frac{\|\phi\|_X^2}{\|\phi_0\|_X^2}$ in semi-logarithmic scale

Simulations were carried out using an *Upwind-type explicit Finite-Difference scheme* (see LeVeque 2002).

OPENING TO PHS FRAMEWORK ?

For *classical solutions*, system (1) can be expressed in *primitive variables* ε_s, v_s .

For simplicity, we omit δ_{z_f} and set $v_m = 0$ in $\mathbf{S}(\mathbf{Y})$. Dynamics may be reformulated by the dissipative Hamiltonian system

$$\frac{\partial}{\partial t} \begin{pmatrix} \varepsilon_s \\ v_s \end{pmatrix} = \begin{pmatrix} 0 & -\frac{\partial}{\partial z} \\ -\frac{\partial}{\partial z} & -\frac{1}{\rho_s} R(\varepsilon_s) \end{pmatrix} \begin{pmatrix} \frac{\delta H}{\delta \varepsilon_s} \\ \frac{\delta H}{\delta v_s} \end{pmatrix},$$

with

$$H(z, \varepsilon_s, v_s) = \int_0^{z_b} \varepsilon_s \frac{v_s^2}{2} dz + \int_0^{\varepsilon_s} \Pi(s) ds - \left(1 - \frac{\rho_l}{\rho_s}\right) g \varepsilon_s z,$$

where

$$\Pi(\varepsilon_s) = \left(\int_{\varepsilon_{\min}}^{\varepsilon_s} \frac{\sigma'_e(s)}{s \rho_s} ds \right) \mathbf{1}_{[\varepsilon_{\min}, \varepsilon_{\max}]}(z),$$

$$R(\varepsilon_s) = \frac{r(\varepsilon_s)}{\varepsilon_s^2 (1 - \varepsilon_s)}.$$

CONCLUSIONS

- Tackling new difficulties related to non-diagonal structure and establishing stability conditions for this type of systems.
- Apply this theory to the Clarifier-Aeration tank configuration

REFERENCES

- Amirat, Mohand Ouidir, Vincent Andrieu, Jean Auriol, Mathieu Bajodek, and C. Valentin (Sept. 2024). "Stability analysis of a class of 2 x 2 triangular hyperbolic systems". In: 4th IFAC Conference of Modelling, Identification and Control of Nonlinear Systems. 7 p. Lyon, France.
- Bastin, Georges and Jean-Michel Coron (2016). *Stability and boundary stabilization of 1-d hyperbolic systems*. V. 88. Springer.
- Bressan, Alberto (2000). *Hyperbolic systems of conservation laws: the one-dimensional Cauchy problem*. V. 20. Oxford University Press, USA.
- Keyfitz, Barbara L and Herbert C Kranzer (1980). "A system of non-strictly hyperbolic conservation laws arising in elasticity theory". In: *Archive for Rational Mechanics and Analysis* 72.3, p. 219-241.
- LeVeque, Randall J (2002). *Finite volume methods for hyperbolic problems*. V. 31. Cambridge university press.